Limit Theorems

Contents

1 Introduction

The most important theoretical results in probability theory are limit theorems. Of these, the most important are those classified either under the heading *laws of large numbers* or under the heading *central limit theorems*. Usually, theorems are considered to be laws of large numbers if they are concerned with stating conditions under which the average of a sequence of random variables converges (in some sense) to the expected average. By contrast, central limit theorems are concerned with determining conditions under which the sum of a large number of random variables has a probability distribution that is approximately normal.

2 Chebyshev's Inequality and the Weak Law of Large Numbers

We start this section by proving a result known as Markov's inequality.

Proposition 2.1 **Markov's inequality**

If X is a random variable that takes only nonnegative values, then for any value $a > 0$,

$$P\{X \geq a\} \leq \frac{E[X]}{a}$$

Proof For $a > 0$, let

$$I = \begin{cases} 1 & \text{if } X \geq a \\ 0 & \text{otherwise} \end{cases}$$

and note that, since $X \geq 0$,

$$I \leq \frac{X}{a}$$

Taking expectations of the preceding inequality yields

$$E[I] \le \frac{E[X]}{a}$$

which, because $E[I] = P\{X \ge a\}$, proves the result. □

As a corollary, we obtain Proposition 2.2.

Proposition 2.2 **Chebyshev's inequality**

If X is a random variable with finite mean μ and variance σ^2, then for any value $k > 0$,

$$P\{|X - \mu| \ge k\} \le \frac{\sigma^2}{k^2}$$

Proof Since $(X - \mu)^2$ is a nonnegative random variable, we can apply Markov's inequality (with $a = k^2$) to obtain

$$P\{(X - \mu)^2 \ge k^2\} \le \frac{E[(X - \mu)^2]}{k^2} \tag{2.1}$$

But since $(X - \mu)^2 \ge k^2$ if and only if $|X - \mu| \ge k$, Equation (2.1) is equivalent to

$$P\{|X - \mu| \ge k\} \le \frac{E[(X - \mu)^2]}{k^2} = \frac{\sigma^2}{k^2}$$

and the proof is complete. □

The importance of Markov's and Chebyshev's inequalities is that they enable us to derive bounds on probabilities when only the mean or both the mean and the variance of the probability distribution are known. Of course, if the actual distribution were known, then the desired probabilities could be computed exactly and we would not need to resort to bounds.

Example 2a Suppose that it is known that the number of items produced in a factory during a week is a random variable with mean 50.

(a) What can be said about the probability that this week's production will exceed 75?

(b) If the variance of a week's production is known to equal 25, then what can be said about the probability that this week's production will be between 40 and 60?

Solution Let X be the number of items that will be produced in a week.

(a) By Markov's inequality,

$$P\{X > 75\} \le \frac{E[X]}{75} = \frac{50}{75} = \frac{2}{3}$$

(b) By Chebyshev's inequality,

$$P\{|X - 50| \ge 10\} \le \frac{\sigma^2}{10^2} = \frac{1}{4}$$

Hence,

$$P\{|X - 50| < 10\} \ge 1 - \frac{1}{4} = \frac{3}{4}$$

so the probability that this week's production will be between 40 and 60 is at least .75. ■

As Chebyshev's inequality is valid for all distributions of the random variable X, we cannot expect the bound on the probability to be very close to the actual probability in most cases. For instance, consider Example 2b.

Example 2b If X is uniformly distributed over the interval (0, 10), then, since $E[X] = 5$ and $\text{Var}(X) = \frac{25}{3}$, it follows from Chebyshev's inequality that

$$P\{|X - 5| > 4\} \leq \frac{25}{3(16)} \approx .52$$

whereas the exact result is

$$P\{|X - 5| > 4\} = .20$$

Thus, although Chebyshev's inequality is correct, the upper bound that it provides is not particularly close to the actual probability.

Similarly, if X is a normal random variable with mean μ and variance σ^2, Chebyshev's inequality states that

$$P\{|X - \mu| > 2\sigma\} \leq \frac{1}{4}$$

whereas the actual probability is given by

$$P\{|X - \mu| > 2\sigma\} = P\left\{\left|\frac{X - \mu}{\sigma}\right| > 2\right\} = 2[1 - \Phi(2)] \approx .0456 \qquad \blacksquare$$

Chebyshev's inequality is often used as a theoretical tool in proving results. This use is illustrated first by Proposition 2.3 and then, most importantly, by the weak law of large numbers.

Proposition 2.3 If $\text{Var}(X) = 0$, then

$$P\{X = E[X]\} = 1$$

In other words, the only random variables having variances equal to 0 are those that are constant with probability 1.

Proof By Chebyshev's inequality, we have, for any $n \geq 1$,

$$P\left\{|X - \mu| > \frac{1}{n}\right\} = 0$$

Letting $n \to \infty$ and using the continuity property of probability yields

$$\begin{aligned} 0 = \lim_{n \to \infty} P\left\{|X - \mu| > \frac{1}{n}\right\} &= P\left\{\lim_{n \to \infty}\left\{|X - \mu| > \frac{1}{n}\right\}\right\} \\ &= P\{X \neq \mu\} \end{aligned}$$

and the result is established. $\square$

Theorem 2.1 **The weak law of large numbers**

Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables, each having finite mean $E[X_i] = \mu$. Then, for any $\varepsilon > 0$,

$$P\left\{\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| \geq \varepsilon\right\} \to 0 \quad \text{as} \quad n \to \infty$$

Proof We shall prove the theorem only under the additional assumption that the random variables have a finite variance σ^2. Now, since

$$E\left[\frac{X_1 + \cdots + X_n}{n}\right] = \mu \quad \text{and} \quad \text{Var}\left(\frac{X_1 + \cdots + X_n}{n}\right) = \frac{\sigma^2}{n}$$

it follows from Chebyshev's inequality that

$$P\left\{\left|\frac{X_1 + \cdots + X_n}{n} - \mu\right| \geq \varepsilon\right\} \leq \frac{\sigma^2}{n\varepsilon^2}$$

and the result is proven. □

The weak law of large numbers was originally proven by James Bernoulli for the special case where the X_i are 0, 1 (that is, Bernoulli) random variables. His statement and proof of this theorem were presented in his book *Ars Conjectandi*, which was published in 1713, eight years after his death, by his nephew Nicholas Bernoulli. Note that because Chebyshev's inequality was not known in Bernoulli's time, Bernoulli had to resort to a quite ingenious proof to establish the result. The general form of the weak law of large numbers presented in Theorem 2.1 was proved by the Russian mathematician Khintchine.

3 The Central Limit Theorem

The central limit theorem is one of the most remarkable results in probability theory. Loosely put, it states that the sum of a large number of independent random variables has a distribution that is approximately normal. Hence, it not only provides a simple method for computing approximate probabilities for sums of independent random variables, but also helps explain the remarkable fact that the empirical frequencies of so many natural populations exhibit bell-shaped (that is, normal) curves.

In its simplest form, the central limit theorem is as follows.

Theorem 3.1 **The central limit theorem**

Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables, each having mean μ and variance σ^2. Then the distribution of

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

tends to the standard normal as $n \to \infty$. That is, for $-\infty < a < \infty$,

$$P\left\{\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \leq a\right\} \to \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{a} e^{-x^2/2}\,dx \quad \text{as} \quad n \to \infty$$

The key to the proof of the central limit theorem is the following lemma, which we state without proof.

Lemma 3.1 Let $Z_1, Z_2, \ldots$ be a sequence of random variables having distribution functions F_{Z_n} and moment generating functions $M_{Z_n}, n \geq 1$, and let Z be a random variable having distribution function F_Z and moment generating function M_Z. If $M_{Z_n}(t) \to M_Z(t)$ for all t, then $F_{Z_n}(t) \to F_Z(t)$ for all t at which $F_Z(t)$ is continuous.

If we let Z be a standard normal random variable, then, since $M_Z(t) = e^{t^2/2}$, it follows from Lemma 3.1 that if $M_{Z_n}(t) \to e^{t^2/2}$ as $n \to \infty$, then $F_{Z_n}(t) \to \Phi(t)$ as $n \to \infty$.

We are now ready to prove the central limit theorem.

Proof of the Central Limit Theorem: Let us assume at first that $\mu = 0$ and $\sigma^2 = 1$. We shall prove the theorem under the assumption that the moment generating function of the $X_i, M(t)$, exists and is finite. Now, the moment generating function of $X_i/\sqrt{n}$ is given by

$$E\left[\exp\left\{\frac{tX_i}{\sqrt{n}}\right\}\right] = M\left(\frac{t}{\sqrt{n}}\right)$$

Thus, the moment generating function of $\sum_{i=1}^{n} X_i/\sqrt{n}$ is given by $\left[M\left(\frac{t}{\sqrt{n}}\right)\right]^n$. Let

$$L(t) = \log M(t)$$

and note that

$$\begin{aligned}
L(0) &= 0 \\
L'(0) &= \frac{M'(0)}{M(0)} \\
&= \mu \\
&= 0 \\
L''(0) &= \frac{M(0)M''(0) - [M'(0)]^2}{[M(0)]^2} \\
&= E[X^2] \\
&= 1
\end{aligned}$$

Now, to prove the theorem, we must show that $[M(t/\sqrt{n})]^n \to e^{t^2/2}$ as $n \to \infty$, or, equivalently, that $nL(t/\sqrt{n}) \to t^2/2$ as $n \to \infty$. To show this, note that

$$\begin{aligned}
\lim_{n\to\infty} \frac{L(t/\sqrt{n})}{n^{-1}} &= \lim_{n\to\infty} \frac{-L'(t/\sqrt{n})n^{-3/2}t}{-2n^{-2}} \quad \text{by L'Hôpital's rule} \\
&= \lim_{n\to\infty} \left[\frac{L'(t/\sqrt{n})t}{2n^{-1/2}}\right] \\
&= \lim_{n\to\infty} \left[\frac{-L''(t/\sqrt{n})n^{-3/2}t^2}{-2n^{-3/2}}\right] \quad \text{again by L'Hôpital's rule} \\
&= \lim_{n\to\infty} \left[L''\left(\frac{t}{\sqrt{n}}\right)\frac{t^2}{2}\right] \\
&= \frac{t^2}{2}
\end{aligned}$$

Thus, the central limit theorem is proven when $\mu = 0$ and $\sigma^2 = 1$. The result now follows in the general case by considering the standardized random variables $X_i^* = (X_i - \mu)/\sigma$ and applying the preceding result, since $E[X_i^*] = 0, \mathrm{Var}(X_i^*) = 1$.

Remark Although Theorem 3.1 states only that, for each a,

$$P\left\{\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \le a\right\} \to \Phi(a)$$

it can, in fact, be shown that the convergence is uniform in a. [We say that $f_n(a) \to f(a)$ uniformly in a if, for each $\varepsilon > 0$, there exists an N such that $|f_n(a) - f(a)| < \varepsilon$ for all a whenever $n \ge N$.] ■

The first version of the central limit theorem was proven by DeMoivre around 1733 for the special case where the X_i are Bernoulli random variables with $p = \frac{1}{2}$. The theorem was subsequently extended by Laplace to the case of arbitrary p. (Since a binomial random variable may be regarded as the sum of n independent and identically distributed Bernoulli random variables, this justifies the normal approximation to the binomial.) Laplace also discovered the more general form of the central limit theorem given in Theorem 3.1. His proof, however, was not completely rigorous and, in fact, cannot easily be made rigorous. A truly rigorous proof of the central limit theorem was first presented by the Russian mathematician Liapounoff in the period 1901–1902.

Figure 1 illustrates the central limit theorem by plotting the probability mass functions of n independent random variables having a specified mass function when (a) $n = 5$, (b) $n = 10$, (c) $n = 25$, and (d) $n = 100$.

Example 3a

An astronomer is interested in measuring the distance, in light-years, from his observatory to a distant star. Although the astronomer has a measuring technique, he knows that because of changing atmospheric conditions and normal error, each time a measurement is made, it will not yield the exact distance, but merely an estimate. As a result, the astronomer plans to make a series of measurements and then use the average value of these measurements as his estimated value of the actual distance. If the astronomer believes that the values of the measurements are independent and identically distributed random variables having a common mean d (the actual

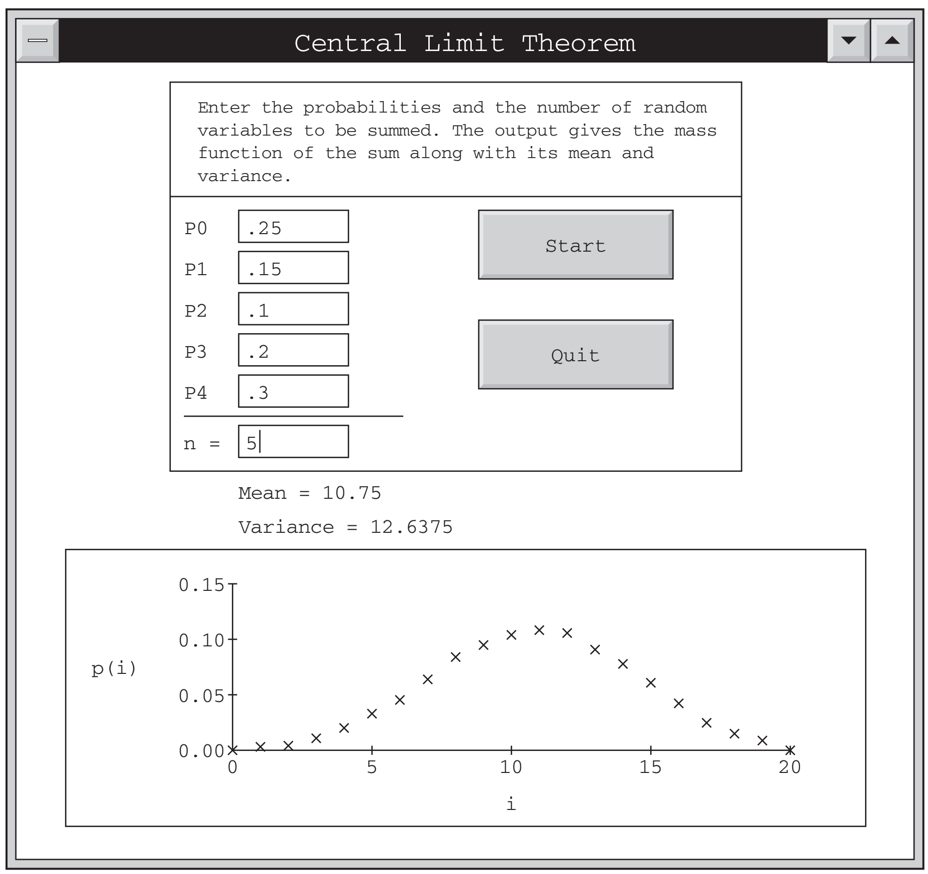

Figure 1(a)

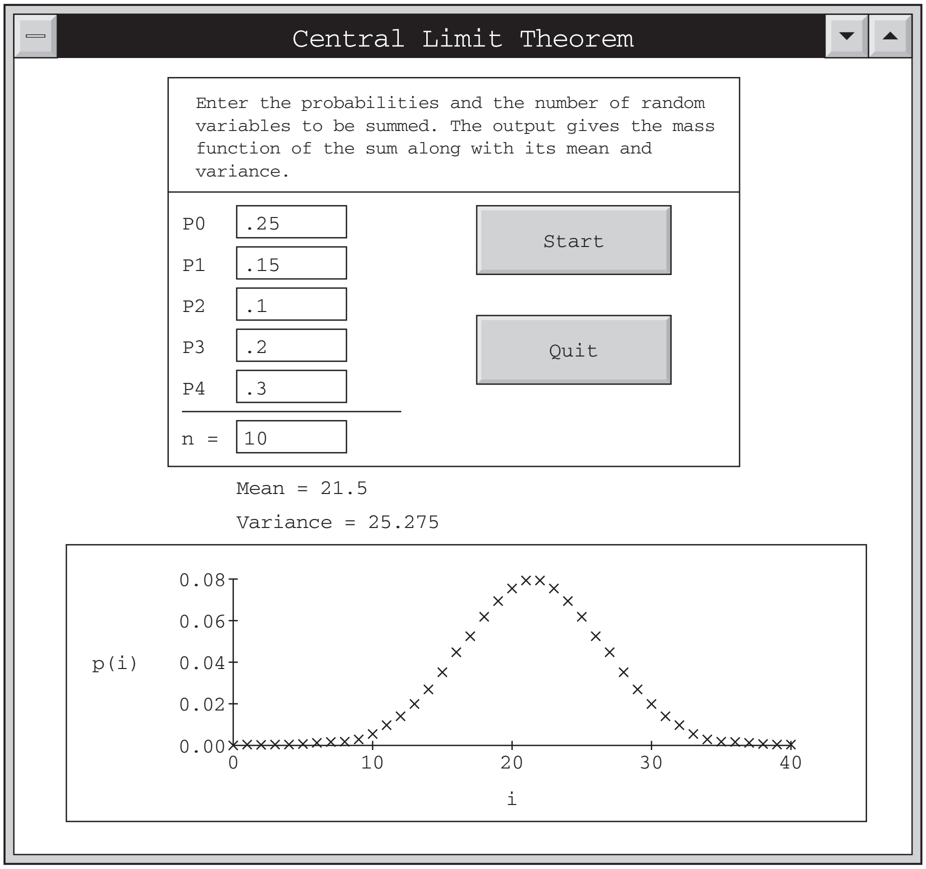

Figure 1(b)

distance) and a common variance of 4 (light-years), how many measurements need he make to be reasonably sure that his estimated distance is accurate to within ±.5 light-year?

Solution Suppose that the astronomer decides to make n observations. If X_1, $X_2, \ldots, X_n$ are the n measurements, then, from the central limit theorem, it follows that

$$Z_n = \frac{\sum_{i=1}^{n} X_i - nd}{2\sqrt{n}}$$

has approximately a standard normal distribution. Hence,

$$P\left\{ -.5 \leq \frac{\sum_{i=1}^{n} X_i}{n} - d \leq .5 \right\} = P\left\{ -.5\frac{\sqrt{n}}{2} \leq Z_n \leq .5\frac{\sqrt{n}}{2} \right\}$$

$$\approx \Phi\left(\frac{\sqrt{n}}{4}\right) - \phi\left(-\frac{\sqrt{n}}{4}\right) = 2\Phi\left(\frac{\sqrt{n}}{4}\right) - 1$$

Therefore, if the astronomer wants, for instance, to be 95 percent certain that his estimated value is accurate to within .5 light-year, he should make n^* measurements, where n^* is such that

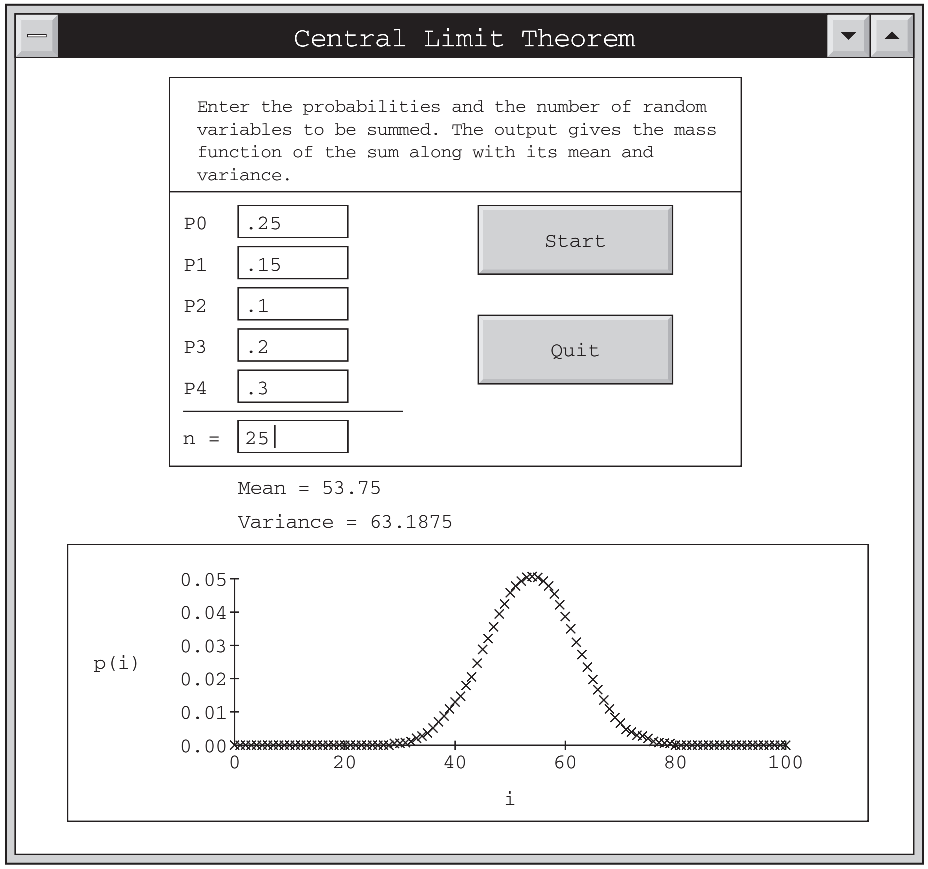

Figure 1(c)

$$2\Phi\left(\frac{\sqrt{n^*}}{4}\right) - 1 = .95 \qquad \text{or} \qquad \Phi\left(\frac{\sqrt{n^*}}{4}\right) = .975$$

Thus,

$$\frac{\sqrt{n^*}}{4} = 1.96 \qquad \text{or} \qquad n^* = (7.84)^2 \approx 61.47$$

As n^* is not integral valued, he should make 62 observations.

Note, however, that the preceding analysis has been done under the assumption that the normal approximation will be a good approximation when $n = 62$. Although this will usually be the case, in general the question of how large n need be before the approximation is "good" depends on the distribution of the X_i. If the astronomer is concerned about this point and wants to take no chances, he can still solve his problem by using Chebyshev's inequality. Since

$$E\left[\sum_{i=1}^{n} \frac{X_i}{n}\right] = d \qquad \text{Var}\left(\sum_{i=1}^{n} \frac{X_i}{n}\right) = \frac{4}{n}$$

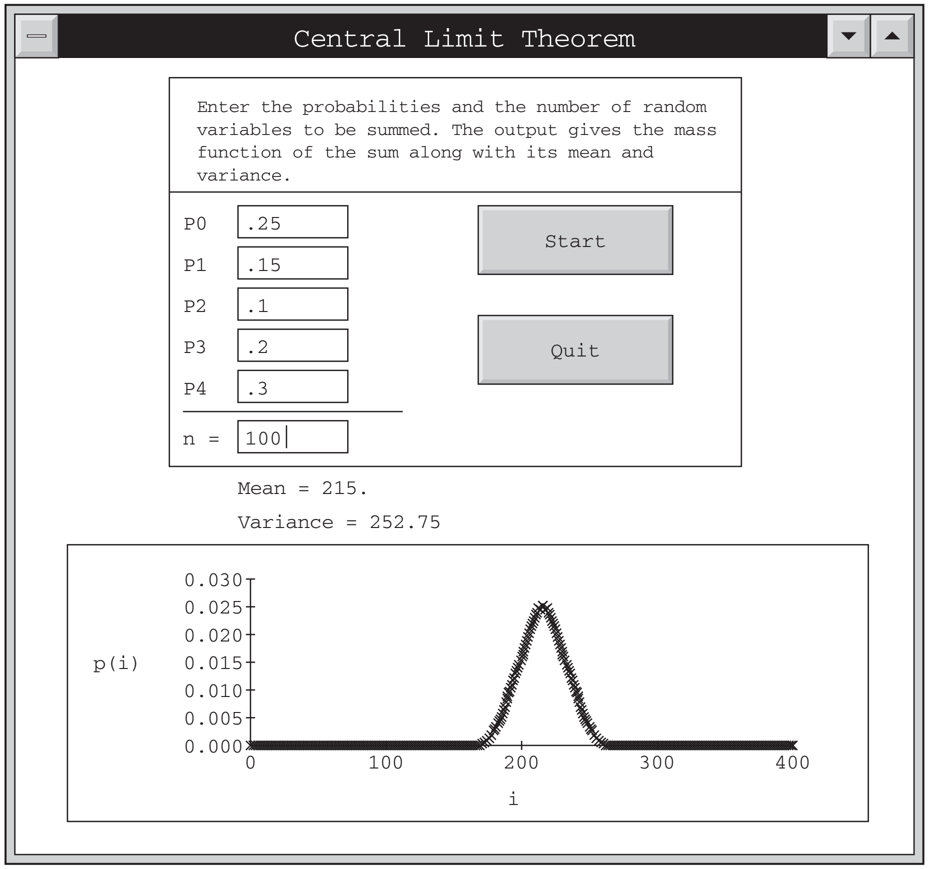

Figure 1(d)

Chebyshev's inequality yields

$$P\left\{\left|\sum_{i=1}^{n}\frac{X_i}{n}-d\right|>.5\right\}\leq\frac{4}{n(.5)^2}=\frac{16}{n}$$

Hence, if he makes $n = 16/.05 = 320$ observations, he can be 95 percent certain that his estimate will be accurate to within .5 light-year. ■

Example 3b The number of students who enroll in a psychology course is a Poisson random variable with mean 100. The professor in charge of the course has decided that if the number enrolling is 120 or more, he will teach the course in two separate sections, whereas if fewer than 120 students enroll, he will teach all of the students together in a single section. What is the probability that the professor will have to teach two sections?

Solution The exact solution

$$e^{-100}\sum_{i=120}^{\infty}\frac{(100)^i}{i!}$$

does not readily yield a numerical answer. However, a Poisson random variable with mean 100 is the sum of 100 independent Poisson random variables, each with mean 1, thus we can make use of the central limit theorem to obtain an approximate solution. If X denotes the number of students who enroll in the course, we have

$$\begin{aligned}P\{X \geq 120\} &= P\{X \geq 119.5\} \quad \text{(the continuity correction)}\\ &= P\left\{\frac{X - 100}{\sqrt{100}} \geq \frac{119.5 - 100}{\sqrt{100}}\right\}\\ &\approx 1 - \Phi(1.95)\\ &\approx .0256\end{aligned}$$

where we have used the fact that the variance of a Poisson random variable is equal to its mean. ■

Example 3c

If 10 fair dice are rolled, find the approximate probability that the sum obtained is between 30 and 40, inclusive.

Solution Let X_i denote the value of the ith die, $i = 1, 2, \ldots, 10$. Since

$$E(X_i) = \frac{7}{2}, \quad \mathrm{Var}(X_i) = E[X_i^2] - (E[X_i])^2 = \frac{35}{12},$$

the central limit theorem yields

$$\begin{aligned}P\{29.5 \leq X \leq 40.5\} &= P\left\{\frac{29.5 - 35}{\sqrt{\frac{350}{12}}} \leq \frac{X - 35}{\sqrt{\frac{350}{12}}} \leq \frac{40.5 - 35}{\sqrt{\frac{350}{12}}}\right\}\\ &\approx 2\Phi(1.0184) - 1\\ &\approx .692\end{aligned}$$

■

Example 3d

Let $X_i, i = 1, \ldots, 10$, be independent random variables, each uniformly distributed over (0, 1). Calculate an approximation to $P\left\{\sum_{i=1}^{10} X_i > 6\right\}$.

Solution Since $E[X_i] = \frac{1}{2}$ and $\mathrm{Var}(X_i) = \frac{1}{12}$, we have, by the central limit theorem,

$$\begin{aligned}P\left\{\sum_1^{10} X_i > 6\right\} &= P\left\{\frac{\sum_1^{10} X_i - 5}{\sqrt{10\left(\frac{1}{12}\right)}} > \frac{6 - 5}{\sqrt{10\left(\frac{1}{12}\right)}}\right\}\\ &\approx 1 - \Phi(\sqrt{1.2})\\ &\approx .1367\end{aligned}$$

Hence, $\sum_{i=1}^{10} X_i$ will be greater than 6 only 14 percent of the time. ■

Example 3e An instructor has 50 exams that will be graded in sequence. The times required to grade the 50 exams are independent, with a common distribution that has mean 20 minutes and standard deviation 4 minutes. Approximate the probability that the instructor will grade at least 25 of the exams in the first 450 minutes of work.

Solution If we let X_i be the time that it takes to grade exam i, then

$$X = \sum_{i=1}^{25} X_i$$

is the time it takes to grade the first 25 exams. Because the instructor will grade at least 25 exams in the first 450 minutes of work if the time it takes to grade the first 25 exams is less than or equal to 450, we see that the desired probability is $P\{X \le 450\}$. To approximate this probability, we use the central limit theorem. Now,

$$E[X] = \sum_{i=1}^{25} E[X_i] = 25(20) = 500$$

and

$$\text{Var}(X) = \sum_{i=1}^{25} \text{Var}(X_i) = 25(16) = 400$$

Consequently, with Z being a standard normal random variable, we have

$$\begin{aligned} P\{X \le 450\} &= P\left\{\frac{X - 500}{\sqrt{400}} \le \frac{450 - 500}{\sqrt{400}}\right\} \\ &\approx P\{Z \le -2.5\} \\ &= P\{Z \ge 2.5\} \\ &= 1 - \Phi(2.5) \approx .006 \end{aligned}$$

■

Central limit theorems also exist when the X_i are independent, but not necessarily identically distributed random variables. One version, by no means the most general, is as follows.

Theorem 3.2 **Central limit theorem for independent random variables**

Let $X_1, X_2, \ldots$ be a sequence of independent random variables having respective means and variances $\mu_i = E[X_i], \sigma_i^2 = \text{Var}(X_i)$. If (a) the X_i are uniformly bounded—that is, if for some $M, P\{|X_i| < M\} = 1$ for all i, and (b) $\sum_{i=1}^{\infty} \sigma_i^2 = \infty$—then

$$P\left\{\frac{\sum_{i=1}^{n}(X_i - \mu_i)}{\sqrt{\sum_{i=1}^{n}\sigma_i^2}} \le a\right\} \to \Phi(a) \quad \text{as} \quad n \to \infty$$

Historical note

Pierre-Simon, Marquis de Laplace

The central limit theorem was originally stated and proven by the French mathematician Pierre-Simon, Marquis de Laplace, who came to the theorem from his observations that errors of measurement (which can usually be regarded as being the sum of a large number of tiny forces) tend to be normally distributed. Laplace, who was also a famous astronomer (and indeed was called "the Newton of France"), was one of the great early contributors to both probability and statistics. Laplace was also a popularizer of the uses of probability in everyday life. He strongly believed in its importance, as is indicated by the following quotations taken from his published book *Analytical Theory of Probability*: "We see that the theory of probability is at bottom only common sense reduced to calculation; it makes us appreciate with exactitude what reasonable minds feel by a sort of instinct, often without being able to account for it.... It is remarkable that this science, which originated in the consideration of games of chance, should become the most important object of human knowledge.... The most important questions of life are, for the most part, really only problems of probability."

The application of the central limit theorem to show that measurement errors are approximately normally distributed is regarded as an important contribution to science. Indeed, in the seventeenth and eighteenth centuries, the central limit theorem was often called the *law of frequency of errors*. Listen to the words of Francis Galton (taken from his book *Natural Inheritance*, published in 1889): "I know of scarcely anything so apt to impress the imagination as the wonderful form of cosmic order expressed by the 'Law of Frequency of Error.' The Law would have been personified by the Greeks and deified, if they had known of it. It reigns with serenity and in complete self-effacement amidst the wildest confusion. The huger the mob and the greater the apparent anarchy, the more perfect is its sway. It is the supreme law of unreason."

4 The Strong Law of Large Numbers

The *strong law of large numbers* is probably the best-known result in probability theory. It states that the average of a sequence of independent random variables having a common distribution will, with probability 1, converge to the mean of that distribution.

Theorem 4.1 **The strong law of large numbers**

Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables, each having a finite mean $\mu = E[X_i]$. Then, with probability 1,

$$\frac{X_1 + X_2 + \cdots + X_n}{n} \to \mu \qquad \text{as} \qquad n \to \infty^{\dagger}$$

As an application of the strong law of large numbers, suppose that a sequence of independent trials of some experiment is performed. Let E be a fixed event of the

† That is, the strong law of large numbers states that

$$P\{\lim_{n \to \infty} (X_1 + \cdots + X_n)/n = \mu\} = 1$$